

Anti-Forgetting Transfer Learning for Cross-Power Fault Diagnosis Based on Continual Learning Strategies

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Abstract—Cross-power fault diagnosis is important for prognostics and health management (PHM) systems because industrial equipment often operates under changing power or load conditions. A diagnostic model trained at one power level may suffer from severe distribution shift when deployed at another power level. Transfer learning can improve target-domain performance, but direct fine-tuning often causes catastrophic forgetting, where source-domain diagnostic knowledge is largely destroyed. This paper studies anti-forgetting transfer learning for a 1.0 kW $\rightarrow$ 3.0 kW fault diagnosis task. A multi-modal CrossViT backbone is used to fuse time-domain vibration windows and time-frequency spectrograms. Based on this backbone, several transfer and continual-learning strategies are systematically compared, including basic fine-tuning, model scaling, layer freezing, elastic weight consolidation (EWC), knowledge distillation, learning without forgetting (LWF), domain-adversarial training, MMD-based distribution alignment, and pseudo-labeling. The experiments show that target-oriented adaptation methods may improve target accuracy but often lose source-domain recognition capability. In contrast, distillation with LWF-style replay provides a better balance between target adaptation and source retention. In the latest stride-128 setting, the Distill+LWF configuration achieves 91.03% source accuracy, 89.10% target accuracy, and a Balance Score of 99.10. Visualization results based on training curves, confusion matrices, and encoder self-attention maps further support the effectiveness of the proposed anti-forgetting strategy.

Index Terms—Fault diagnosis, transfer learning, continual learning, catastrophic forgetting, knowledge distillation, domain adaptation, PHM.

I. INTRODUCTION

Rotating machinery and bearing systems are widely used in industrial equipment, transportation systems, and aerospace devices. Their unexpected degradation may cause production loss, safety risks, and expensive maintenance. PHM techniques aim to monitor equipment condition, identify fault patterns, and support maintenance decisions. With the development of sensors and data-driven modeling, deep learning has become a common tool for intelligent fault diagnosis because it can automatically extract discriminative features from vibration signals. Recent PHM and industrial monitoring studies also show that deep models and large-scale visual-language models can enhance health assessment and intelligent inspection under complex industrial environments [1], [2].

A key assumption behind many supervised deep diagnostic models is that the training and testing samples follow similar distributions. This assumption is frequently violated in practical scenarios. When the equipment power, speed, or load changes, the vibration amplitude, frequency content, and time-frequency structures also change. Consequently, a model trained under one condition may fail under another condition. Existing transfer diagnosis studies have shown that domain adaptation and transfer learning can reduce cross-domain distribution discrepancy and improve target-domain recognition performance [3]–[7].

However, most cross-domain fault diagnosis methods mainly evaluate target-domain performance. In many engineering applications, the equipment may switch between several power conditions during operation. Therefore, a reliable model should not only adapt to the new target domain, but also retain source-domain diagnostic knowledge. Direct fine-tuning usually improves the target domain by updating the entire model toward the target distribution, but the decision boundary learned from the source domain is easily forgotten. This catastrophic forgetting problem limits the deployment of transfer diagnosis systems in multi-condition PHM tasks.

Continual learning offers a useful perspective for this problem. Methods such as EWC, knowledge distillation, and LWF attempt to preserve previous-task knowledge while learning new tasks [8]–[10]. Motivated by these methods, this paper studies cross-power fault diagnosis as a dual-objective transfer problem. The objective is not only high target-domain accuracy, but also stable source-domain retention. The main contributions are as follows.

First, a multi-modal CrossViT diagnosis model is constructed for cross-power fault diagnosis by combining time-domain signal encoding, spectrogram encoding, and cross-modal attention fusion. Second, a systematic transfer-learning pipeline is developed from EXP1 to EXP3 V4, covering fine-tuning, model scaling, freezing, EWC, distillation, LWF, DANN, MMD, and pseudo-labeling. Third, the paper introduces a dual-domain evaluation protocol, where source retention, target accuracy, and Balance Score are jointly considered. The latest experiments show that Distill+LWF gives

the strongest overall performance.

II. RELATED WORK

A. Deep Learning and Multi-Modal Fault Diagnosis

Deep learning-based fault diagnosis has been widely studied for rotating machinery. CNN-based and transformer-based models can learn local and long-range signal structures in an end-to-end manner. Time-frequency representations, such as STFT and wavelet-based maps, are also frequently used because they reveal frequency and energy distributions that may not be obvious in raw time signals. Recent work has further combined convolutional modules and attention mechanisms to extract local and global representations for bearing diagnosis [11]. Dual-domain signal fusion methods also indicate that time-domain and frequency-domain signals provide complementary information for cross-domain diagnosis [6].

B. Transfer Learning and Domain Adaptation

Transfer learning uses source-domain knowledge to improve target-domain learning. In fault diagnosis, the source domain usually contains sufficient labeled samples, while the target domain may contain limited or unlabeled samples under another machine, load, or power condition. Domain adaptation methods reduce feature discrepancy between domains through distribution matching or adversarial learning. DSCNN and CAN minimize MK-MMD to learn transferable features [3], [4], [12]. DANN introduces a domain discriminator and gradient reversal layer to learn domain-invariant representations [13]. Other methods jointly align marginal and conditional distributions, or use Wasserstein distance and adversarial training to improve transfer performance [11], [14], [15].

Although these methods are effective for target-domain adaptation, they do not always preserve source-domain performance. Distribution alignment may push the representation toward domain invariance, but domain invariance alone cannot guarantee old decision-boundary retention. This motivates the use of continual-learning strategies in cross-power diagnosis.

C. Continual Learning and Knowledge Retention

Continual learning studies how a model can learn new tasks without forgetting old ones. EWC protects important parameters by using a Fisher-information-weighted penalty [8]. Knowledge distillation transfers the softened output distribution of a teacher model to a student model [9]. LWF constrains the new model to preserve old-task responses during new-task learning [10]. In this paper, these strategies are adapted to cross-power fault diagnosis. The source-domain model is regarded as a teacher, and source replay is used to maintain source decision boundaries while learning target-domain features.

III. METHOD

A. Problem Definition and Multi-Modal Construction

Let $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ denote the labeled source domain collected under 1.0 kW, and $D_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ denote the target domain collected under 3.0 kW. The label space is

shared across domains, but the feature distributions differ due to power variation, i.e., $P_s(x) \neq P_t(x)$. Each raw sample is a three-channel vibration signal. The signal is segmented into fixed-length time windows and transformed into a time-frequency spectrogram by STFT. Thus, each sample has two modalities: a time-domain input x_{time} and a spectrogram input x_{spec} .

B. Multi-Modal CrossViT Backbone

The backbone contains a time-domain encoder, a spectrogram encoder, a cross-modal attention fusion module, and a classifier head. The time-domain encoder projects the segmented waveform into a token sequence and extracts temporal dependencies. The spectrogram encoder uses two-dimensional patch embedding and transformer blocks to extract frequency and time-frequency structures. The two representations are then fused through cross-modal attention:

$$F_t = E_t(x_{time}), \quad F_s = E_s(x_{spec}), \quad (1)$$

$$F_{fusion} = \mathcal{A}_{cross}(F_t, F_s), \quad \hat{y} = C(F_{fusion}), \quad (2)$$

where E_t and E_s are the two encoders, $\mathcal{A}_{cross}(\cdot)$ denotes cross-modal attention fusion, and $C(\cdot)$ is the classifier.

C. Source-Domain Pre-Training

The first stage trains the multi-modal CrossViT on the source domain. The source loss is the standard cross-entropy loss:

$$\mathcal{L}_{src} = -\frac{1}{B} \sum_{i=1}^B \sum_{c=1}^C \mathbf{1}(y_i = c) \log p_i(c). \quad (3)$$

The trained model is saved as a teacher model f_{θ^*} and also provides initialization for target adaptation.

D. Anti-Forgetting Transfer With Distillation, LWF and Freezing

During target adaptation, direct fine-tuning can overwrite source-domain representations. To reduce forgetting, the proposed strategy combines three mechanisms. First, layer freezing is optionally used to reduce large parameter drift in early encoders. Second, a teacher model provides soft labels. Third, source replay following the LWF idea mixes a small ratio of source-domain samples with target-domain samples.

The knowledge distillation loss is

$$\mathcal{L}_{KD} = T^2 \text{KL}(\sigma(z_s/T), \sigma(z_T/T)), \quad (4)$$

where z_s and z_T are student and teacher logits, T is the temperature, and $\sigma(\cdot)$ is softmax. The final loss is

$$\mathcal{L} = (1 - \alpha)\mathcal{L}_{CE} + \alpha\mathcal{L}_{KD} + \lambda_e\mathcal{L}_{EWC}, \quad (5)$$

where α controls the distillation weight and $\mathcal{L}_{EWC}$ is optional.

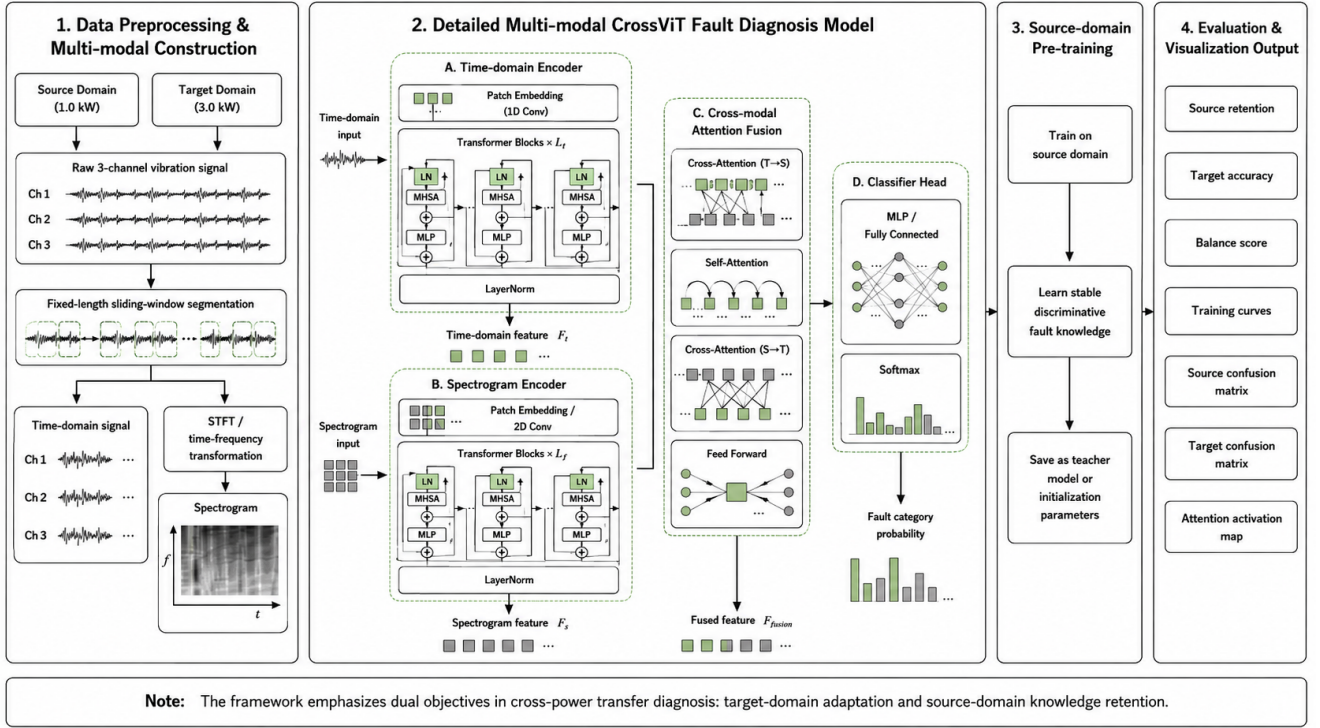


Fig. 1: Overall framework of the proposed anti-forgetting cross-power fault diagnosis method. The framework consists of multi-modal construction, a detailed CrossViT diagnosis backbone, source-domain pre-training, and visualization-based evaluation.

Algorithm 1: Multi-Modal Source-Domain Pre-Training

Input: source domain D_s , model f_θ , maximum epochs E_s .
Output: teacher model f_{θ^*} and initialization parameters.

- 1: Initialize time encoder, spectrogram encoder, fusion module and classifier.
- 2: **for** epoch = 1, ..., E_s **do**
- 3: Sample a mini-batch $(x_{time}^s, x_{spec}^s, y^s)$ from D_s .
- 4: Compute $F_t = E_t(x_{time}^s)$ and $F_s = E_s(x_{spec}^s)$.
- 5: Fuse features by $F_{fusion} = \mathcal{A}_{cross}(F_t, F_s)$.
- 6: Predict $\hat{y}^s = C(F_{fusion})$ and compute $\mathcal{L}_{src}$.
- 7: Update model parameters by back-propagation.
- 8: **end for**
- 9: Save the best source-domain model as teacher f_{θ^*} .

Fig. 2: Source-domain pre-training for stable fault knowledge learning.

E. Domain Adaptation Candidates and Model Selection

DANN, MMD and pseudo-labeling are treated as ablation candidates. DANN uses a domain discriminator connected by a gradient reversal layer. MMD minimizes feature-distribution discrepancy. Pseudo-labeling selects high-confidence target predictions as temporary labels. Since these methods may improve target-domain accuracy while damaging source retention, the final model is selected by a dual-domain Balance Score:

$$S_{bal} = A_t - \beta \max(0, A_s^{ref} - A_s), \quad (6)$$

where A_t is target accuracy, A_s is source accuracy, and A_s^{ref} is a source retention reference. A higher S_{bal} indicates a better

Algorithm 2: Anti-Forgetting Transfer With Freezing, EWC, Distillation and LWF

Input: teacher f_{θ^*} , source replay set R_s , target set D_t , temperature T , weight α .
Output: adapted model $f_{\theta'}$.

- 1: Initialize student $f_\theta \leftarrow f_{\theta^*}$.
- 2: Freeze selected embedding or encoder layers if required.
- 3: Estimate Fisher matrix on D_s when EWC is enabled.
- 4: **for** epoch = 1, ..., E_t **do**
- 5: Sample target mini-batch B_t and replay mini-batch $B_s \subset R_s$.
- 6: Compute student logits z_s and teacher logits z_T .
- 7: Compute hard-label loss $\mathcal{L}_{CE}$ on available labels.
- 8: Compute distillation loss $\mathcal{L}_{KD}$ from softened teacher outputs.
- 9: Add optional $\mathcal{L}_{EWC}$ to protect important parameters.
- 10: Optimize $\mathcal{L} = (1 - \alpha)\mathcal{L}_{CE} + \alpha\mathcal{L}_{KD} + \lambda_e\mathcal{L}_{EWC}$.
- 11: **end for**
- 12: Return the model with the best Balance Score.

Fig. 3: Anti-forgetting transfer procedure used in the final recommended pipeline.

balance.

IV. EXPERIMENTS

A. Dataset and Experimental Settings

The experiments are conducted on a cross-power vibration diagnosis task from 1.0 kW to 3.0 kW. Each domain contains three-channel vibration signals. The model uses fixed-length time windows of length 1024 and spectrograms of size $3 \times$

Algorithm 3: Ablation Candidates and Dual-Domain Model Selection

Input: candidate set $\mathcal{M}$, source test set, target test set.

Output: selected model m^* .

- 1: Construct candidates: fine-tuning, freezing, EWC, Distill, Distill+LWF, DANN, MMD and pseudo-labeling.
 - 2: **for each** candidate $m \in \mathcal{M}$ **do**
 - 3: Train or adapt m under the same source-target task.
 - 4: Evaluate source retention $A_s(m)$ and target accuracy $A_t(m)$.
 - 5: Compute Balance Score $S_{bal}(m)$.
 - 6: **end for**
 - 7: Select $m^* = \arg \max_m S_{bal}(m)$.
 - 8: Generate confusion matrices, training curves and attention maps for m^* .
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Fig. 4: Ablation-driven model selection for dual-objective transfer diagnosis.

TABLE I: Dataset Settings for Cross-Power Fault Diagnosis

Item	Source domain	Target domain
Power condition	1.0 kW	3.0 kW
Signal type	3-channel vibration	3-channel vibration
Time window	1024×3	1024×3
Spectrogram	$3 \times 128 \times 128$	$3 \times 128 \times 128$
Fault categories	16	16

128×128 . The source and target domains share the same 16 fault categories.

B. Overall Quantitative Results

Table II summarizes the complete experimental evolution. The observation column has been removed to keep the configuration field readable. The results include the original stride-256 series and the latest stride-128 series. EXP1 gives the single-domain baseline. EXP2 verifies that direct cross-power generalization fails. EXP3 V1 to V4 then gradually introduce fine-tuning, model scaling, freezing, continual-learning strategies, and domain adaptation candidates.

The progression in Table II shows that a naive cross-power model is not reliable. EXP2 obtains high source-domain accuracy but almost fails in the target domain. V1 confirms that basic fine-tuning is not sufficient. V2 improves target-domain performance through model scaling and freezing, yet source-domain retention remains very low, indicating severe catastrophic forgetting.

The V3 and V4 results further show the difference between target-oriented adaptation and anti-forgetting adaptation. Direct fine-tuning and domain-alignment candidates such as DANN and MMD can raise target-domain accuracy, but they often damage the source-domain decision boundary. EWC and simple freezing are more conservative, but they restrict target-domain adaptation and therefore do not provide the best overall trade-off.

The strongest finding is that knowledge distillation combined with LWF-style source replay gives the most stable dual-domain behavior. In the latest stride-128 setting, Distill+LWF with $T = 2$ and $\alpha = 0.3$ achieves 91.03% source accuracy, 89.10% target accuracy, and a Balance Score of 99.10. The lat-

est V4 distillation setting with $T = 1.5$ and $\alpha = 0.25$ obtains the highest source retention, 94.23%, while still maintaining 87.18% target accuracy. Therefore, Table II is sufficient to support the main quantitative conclusion, and detailed ablation results are discussed in text rather than using another dense table.

More specifically, freezing only early layers protects a small part of the representation but restricts the feature shift required by the target power condition. EWC introduces a Fisher-weighted constraint on important parameters, yet its performance is sensitive to the regularization weight and does not consistently prevent forgetting. DANN and MMD are useful as target-oriented domain adaptation candidates, but their low source-domain retention indicates that domain invariance alone is not equivalent to continual reliability. Pseudo-label learning is the least stable strategy in this task because the initial target predictions are not confident enough under the large power-induced distribution gap.

For this reason, the paper treats DANN, MMD and pseudo-label learning as ablation candidates rather than the final recommended solution. The final recommendation is a distillation-and-replay strategy: the teacher model preserves the source-domain soft decision structure, while replayed source windows prevent the classifier head from drifting completely toward the target domain.

C. Visualization Analysis

The following figures are placed after the quantitative discussion rather than immediately after the tables. This separates numerical comparison from visual evidence. The confusion matrices highlight class-level source retention and target adaptation, while the training curves show the convergence behavior of the selected distillation configurations. In addition, an encoder self-attention map is visualized for a representative correctly classified sample. Since the implemented cross-modal fusion operates after global feature pooling, post-pooling cross-attention weights describe modality-level fusion rather than token-level localization. Therefore, the attention analysis focuses on the self-attention maps inside the time-domain and spectrogram encoders, where token-level temporal and spectral dependencies are explicitly modeled.

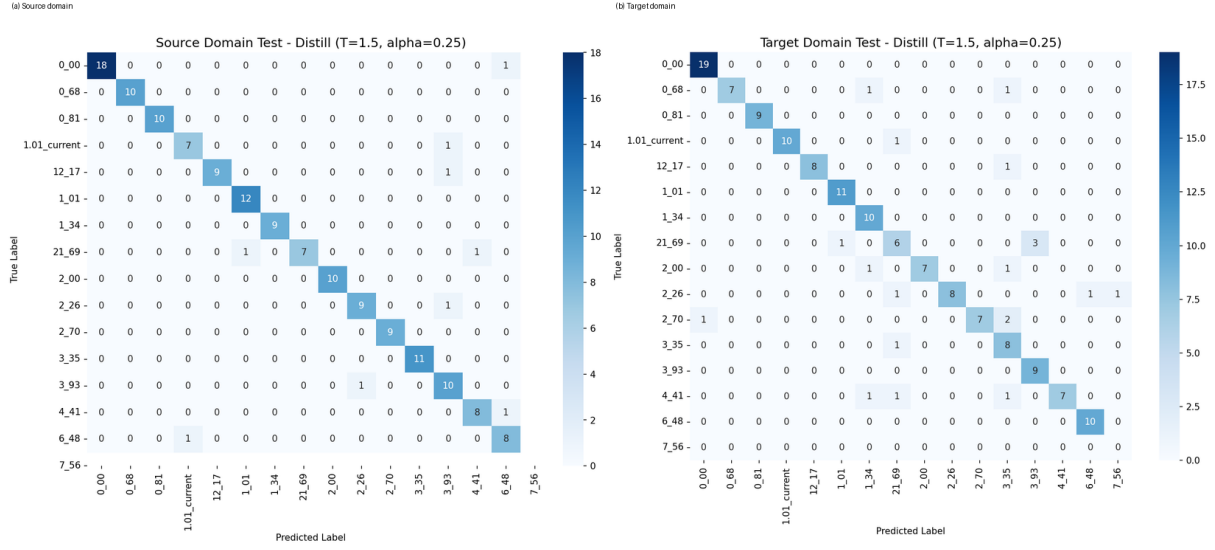
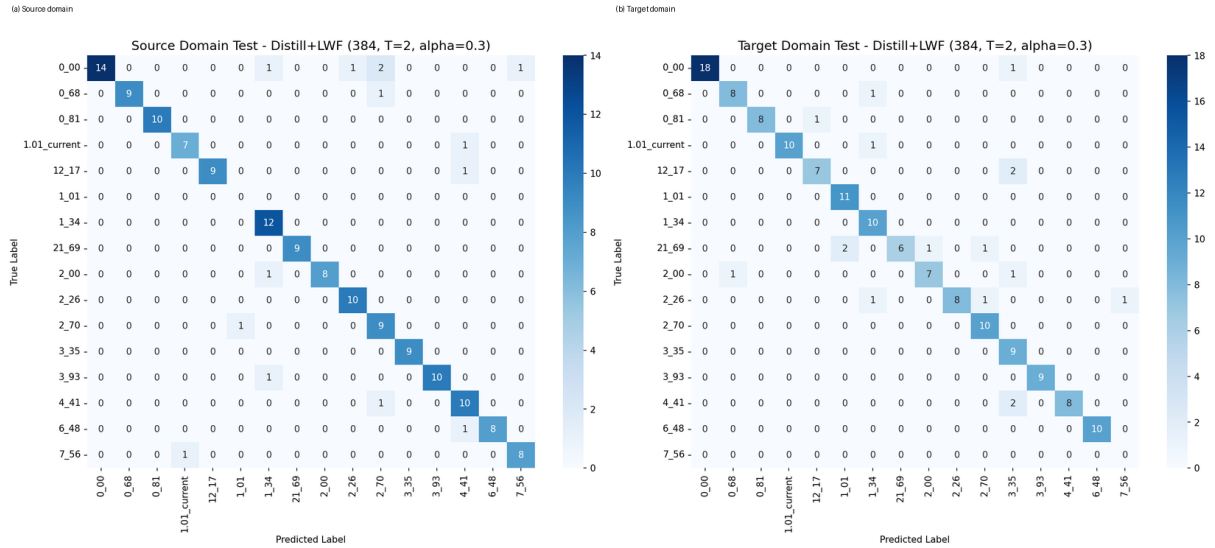
Fig. 5 corresponds to the latest V4 configuration. The two matrices show that the model keeps a clear diagonal structure in both domains, which means the distillation objective does not simply overfit the target domain.

Fig. 6 corresponds to the latest V3 Distill+LWF configuration, which obtains the highest Balance Score. Compared with target-only adaptation, the source-domain confusion matrix remains much more diagonal, demonstrating the effect of replay and teacher constraints. The target-domain matrix also keeps most samples around the diagonal, indicating that anti-forgetting does not simply sacrifice adaptation for retention.

Fig. 7 shows stable convergence during both source pre-training and target fine-tuning. The gap between training and validation accuracy is moderate in the distillation-based set-

TABLE II: Evolution of EXP1, EXP2, EXP3 V1-V4 and Latest Stride-128 Results

Stage	Setting / Best configuration	Source (%)	Target (%)	Balance
EXP1	1.0 kW single-domain baseline	87.82 / 90.00	—	—
EXP2	1.0 kW $\rightarrow$ 3.0 kW, no adaptation	94.23 / 85.00	4.49 / 10.00	—
EXP3 V1	Basic fine-tuning	12.50	27.50	27.50
EXP3 V2	Model scaling + freezing	5.00	67.50	67.50
EXP3 V3	Distill+LWF, $T = 2$, $\alpha = 0.3$	77.50	65.00	75.00
EXP3 V4	Distill, $T = 1.8$, $\alpha = 0.3$	60.00	70.00	80.00
Latest V3	Distill+LWF, $T = 2$, $\alpha = 0.3$	91.03	89.10	99.10
Latest V4	Distill, $T = 1.5$, $\alpha = 0.25$	94.23	87.18	97.18

Fig. 5: Source-domain and target-domain confusion matrices of Distill ($T = 1.5$, $\alpha = 0.25$).Fig. 6: Source-domain and target-domain confusion matrices of Distill+LWF ($T = 2$, $\alpha = 0.3$).

tings, indicating that the teacher constraint and replay samples help stabilize adaptation.

Fig. 8 shows a high-confidence correctly classified target-domain sample. The time-domain attention heads contain localized and banded structures, indicating that the model

captures dependencies among neighboring signal patches. The spectrogram attention heads exhibit structured vertical or block-like responses over selected spectral tokens, suggesting that the frequency branch emphasizes discriminative time-

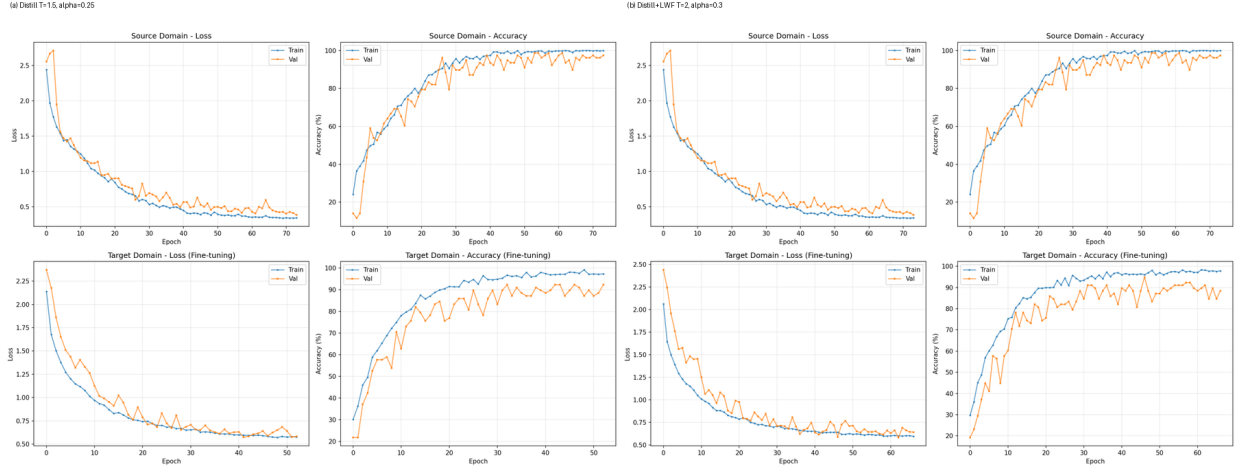


Fig. 7: Training curves of representative distillation-based configurations.

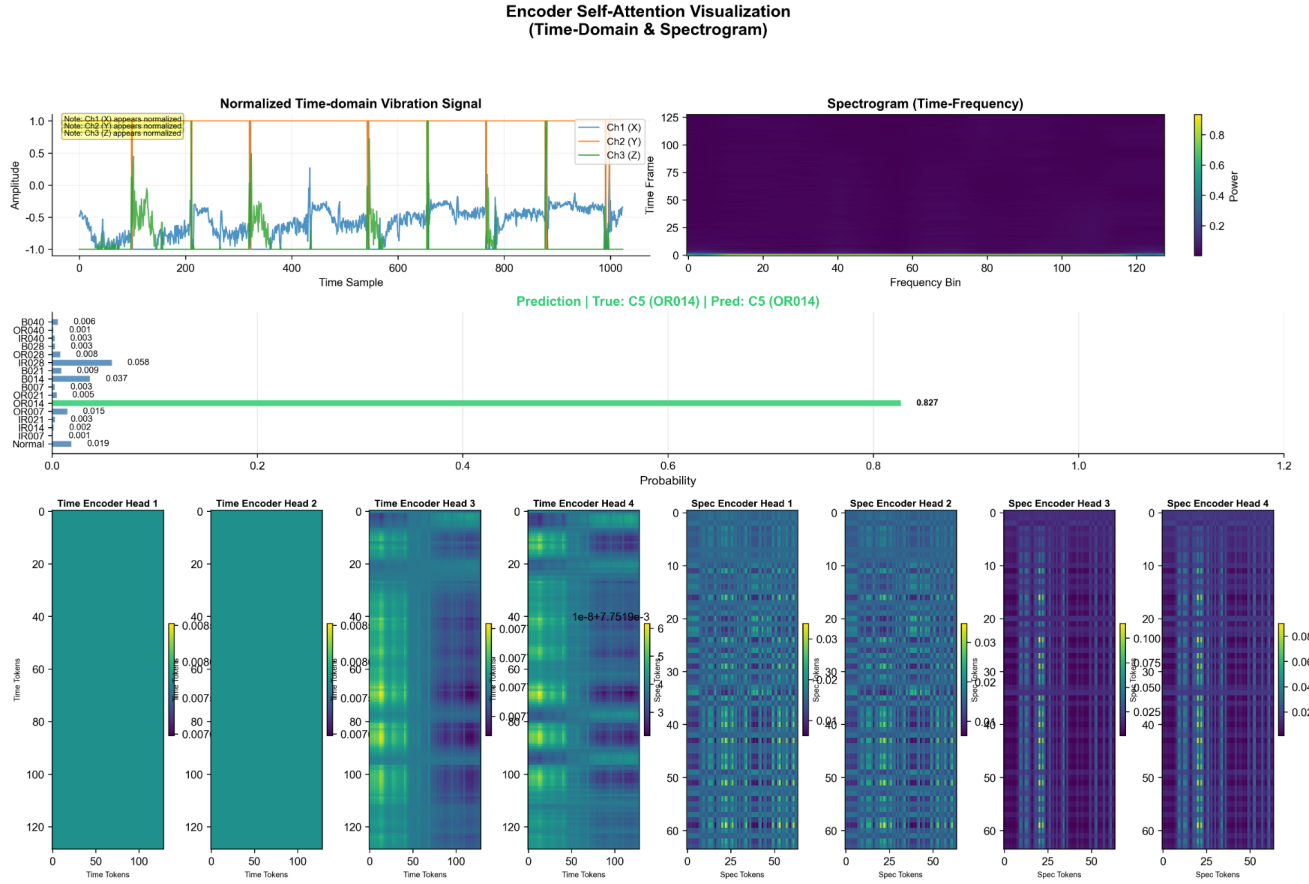


Fig. 8: Encoder self-attention visualization for a correctly classified C5 sample. The upper part shows the normalized three-channel vibration signal, spectrogram and prediction probability, while the lower part visualizes representative self-attention heads from the time-domain and spectrogram encoders.

frequency regions rather than uniformly attending to the whole spectrogram.

V. CONCLUSION

This paper reformulates cross-power fault diagnosis as a dual-objective transfer problem that evaluates both target adaptation and source-domain retention. A multi-modal CrossViT

backbone is combined with continual-learning strategies and tested from EXP1 to EXP3 V4. The results show that pure fine-tuning, DANN and MMD can improve target-domain accuracy but usually cause severe source-domain forgetting. In contrast, knowledge distillation with LWF gives the most reliable balance. In the latest stride-128 setting, Distill+LWF achieves 91.03% source accuracy, 89.10% target accuracy and

a Balance Score of 99.10. Future work will consider multi-source transfer and online continual adaptation.

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